Executive Summary

Nassau Candy Distributor – Factory Reallocation & Shipping Optimization

This project analyzes sales, profitability, product performance, regional distribution, and shipping operations using the Nassau Candy Distributor dataset. The objective is to identify opportunities to improve operational efficiency, optimize shipping processes, and support data-driven resource allocation.

The dataset contains **10,194 records and 18 columns**, including order dates, ship dates, shipping modes, customer locations, product details, sales, units, gross profit, and cost. SQL and Power BI are used to examine key performance indicators, regional sales patterns, product profitability, and order-to-shipment lead times.

The analysis is intended to support evidence-based decisions by identifying potential operational bottlenecks, highlighting regional and product-level performance differences, and establishing measurable indicators for evaluating proposed improvements. Key proposed indicators include lead time reduction, profit impact stability, scenario confidence, and recommendation coverage.

For government stakeholders, the project offers a data-driven framework that could inform logistics planning, resource allocation, operational monitoring, and transparent performance reporting. However, factory reallocation benefits and shipping improvements must be validated through scenario analysis before implementation. The available data measures the time between order placement and shipment, not final delivery time, and does not independently establish actual factory capacity or allocation constraints.

Expected outcome: A Power BI dashboard and analytical framework to help stakeholders monitor performance, assess potential optimization scenarios, and make informed operational decisions based on measurable evidence.